The Hilbert Substrate Framework II: Accessibility and the Emergence of Spatial Locality

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Abstract

Classical spatial structure persists in physical systems despite underlying unitary dynamics that, in principle, permit the widespread delocalization of information. In this work, we investigate whether spatial locality arises as a fundamental optimum or as a dynamical attractor. By applying a locality-biased Riemannian flow to unitary orbits, we demonstrate the existence of an *accessibility phase transition.* For small systems ($N \leq 3$) or weak locality penalties ($p \leq 2$), optimization flows easily access the global minimum corresponding to delocalized collective modes ("Harmonions"). However, beyond a critical system size ($N \geq 5$) and penalty strength ($p \approx 4$), the accessibility landscape undergoes a qualitative transition: the global minimum becomes kinetically inaccessible, and the system is trapped in a robust local basin corresponding to classical spatial geometry. We map the phase diagram of this transition and conclude that spatial locality emerges not because it is the simplest description of physics, but because it is the simplest *accessible* description for large-scale systems.

1 Introduction

1.1 The Problem of Persistent Structure

The standard formulation of quantum mechanics presupposes a tensor product structure of Hilbert space, a notion of locality, and a background spacetime in which physics unfolds. However, these features are not guaranteed by the axioms of quantum theory alone. Under generic unitary evolution, information tends to delocalize rapidly, scrambling local correlations into global entanglement. From this perspective, the persistence of classical spatial structure—the fact that the universe continues to appear as a collection of distinguishable, local subsystems rather than a featureless quantum equilibrium—presents a non-trivial foundational problem.

In previous work, we introduced the *Hilbert Substrate Framework* and the diagnostic of *Heterogeneity in Information Propagation (HIP)* to characterize how structure persists in information flow [1]. While that work established that heterogeneous transport is possible under minimal constraints, it did not explain *why* the specific structure of spatial locality is selected over other possible tensor factorizations. The central question remains: why does the universe organize itself into a geometric lattice rather than, for instance, a basis of delocalized momentum modes?

1.2 Optimality versus Accessibility

The central hypothesis of this work is that the structure of physical reality is determined not by *optimality*, but by *accessibility*.

Ideally, the "simplest" description of an isolated quantum system is its energy eigenbasis (which we term the "Harmonion" basis). In this basis, the Hamiltonian is diagonal, and dynamics are trivial phase rotations. If the universe were optimized solely for dynamical simplicity, we should experience a reality composed of independent, delocalized modes.

However, we propose that for sufficiently large interacting systems, this global optimum becomes dynamically inaccessible. We view the manifold of unitary orbits as a complex energy landscape defined by a "Locality Cost" functional. We hypothesize that as the system size (N) increases, this landscape undergoes a topological transition—analogous to the kinetic arrest observed in spin glasses or protein folding—where the deep global minimum (the eigenbasis) becomes kinetically isolated. In this high-complexity regime, broad, robust local minima corresponding to spatial geometry become the dominant attractors.

1.3 Summary of Results

To test this hypothesis, we perform a series of numerical experiments on unitary orbits, applying a Riemannian optimization flow to minimize locality cost starting from random, scrambled states. Our results identify an *accessibility phase transition* governed by system size and the strictness of the locality constraint:

- **The Quantum Fluid Phase ($N \leq 3$):** For small systems, the optimization landscape is smooth. The flow reliably locates the global minimum (the Harmonion basis), suggesting that spatial geometry cannot exist as a stable distinction at the micro-scale.
- **The Accessibility Collapse ($N \geq 5$):** Beyond a critical system size, the global minimum becomes effectively walled off. The optimization flow consistently converges to a sub-optimal but robust basin of attraction corresponding to 1D spatial geometry.
- **The Phase Diagram:** By varying the strength of the locality penalty (p), we map a phase diagram revealing that spatial geometry emerges only within a specific "window of reality" ($p \approx 3-4$), bounded by quantum delocalization on one side and glassy disorder on the other.

These findings suggest that space is not a fundamental background, but a dynamical attractor—a "good enough" solution that the universe settles into because the perfect solution is out of reach.

2 Formalism

2.1 The Unitary Orbit and Basis Selection

We consider a closed quantum system described by a finite-dimensional Hilbert space $\mathcal{H} \cong (\mathbb{C}^d)^{\otimes N}$, representing N qudits (here, qubits with $d = 2$). The physics of the system is encoded in its Hamiltonian H.

From the perspective of pure unitary dynamics, the specific tensor factorization of $\mathcal{H}$ is not fixed. Two Hamiltonians H and H' are physically equivalent if they are related by a unitary transformation:

$$H' = UHU^\dagger, \quad U \in SU(d^N). \tag{1}$$

Such Hamiltonians share an identical spectrum and partition function, meaning they are indistinguishable via thermodynamic measurements of the closed system. However, they differ in their *structure* relative to a fixed local basis.

We posit a fixed reference decomposition of Hilbert space (the "substrate") defined by the N-qubit Pauli basis $\mathcal{P} = \{\sigma_1^\alpha \otimes \cdots \otimes \sigma_N^\gamma\}$. Our goal is to determine which representative H' on the unitary orbit of H minimizes structural complexity relative to this substrate.

2.2 The Locality Cost Functional

To quantify structural complexity, we introduce a *Locality Cost* functional $\mathcal{C}_p(H)$. We expand the Hamiltonian in the Pauli basis:

$$H = \sum_{k=1}^{4^N} c_k P_k, \quad c_k = \frac{1}{2^N} \operatorname{Tr}(H P_k). \tag{2}$$

We assign a locality weight $w(P_k)$ to each basis element, equal to its Hamming weight (the number of non-identity factors). The cost functional is defined as the p-norm weighted average of these weights:

$$\mathcal{C}_p(H) = \frac{\sum_k w(P_k)^p \, |c_k|^2}{\sum_k |c_k|^2}. \tag{3}$$

Here, $p \geq 1$ is a control parameter representing the "metric pressure" applied to the system.

- For $p = 1$, the cost measures the average locality.
- As $p \to \infty$, the cost is dominated by the most non-local term in the expansion (the ∞-norm).

Minimizing $\mathcal{C}_p(H)$ is equivalent to finding the basis in which the Hamiltonian looks "most local."

2.3 Riemannian Optimization Flow

To explore the landscape of structural complexity, we treat the unitary orbit $\mathcal{O}_H = \{UHU^\dagger \mid U \in SU(d^N)\}$ as a Riemannian manifold. We seek a trajectory $H(t)$ that minimizes $\mathcal{C}_p(H(t))$.

The Euclidean gradient of the cost function with respect to the Hamiltonian coefficients is given by the operator M:

$$M = \sum_k \frac{\partial \mathcal{C}_p}{\partial c_k} P_k = \frac{2}{\sum |c_j|^2} \sum_k w(P_k)^p c_k P_k. \tag{4}$$

Projecting this gradient onto the tangent space of the unitary orbit yields the *Double Bracket Flow* (or Brockett flow) equation [2]:

$$\frac{dH}{dt} = [H, [H, M]]. \tag{5}$$

This flow has two crucial properties:

1. **Isospectrality:** The flow is generated by the unitary commutator $K = [H, M]$, ensuring that the spectrum of $H(t)$ remains constant.
2. **Monotonic Descent:** The cost function decreases monotonically, $\frac{d}{dt}\mathcal{C}_p \leq 0$, with fixed points occurring only when $[H, M] = 0$.

In our numerical experiments, we implement a discrete-time approximation of this flow using Riemannian gradient descent with backtracking line search.

We emphasize that this flow is not intended as a physical time evolution, but as a diagnostic probe of accessibility on the unitary orbit.

2.4 Reference Structures

To interpret the results of the flow, we define two reference cost values for any given system:

1. The Spatial Cost ($\mathcal{C}_{\text{spatial}}$): The cost of the Hamiltonian in the standard spatial basis (e.g., nearest-neighbor interactions). For a 1D Heisenberg chain, $\mathcal{C}_{\text{spatial}} = 2^p$ (since all terms have weight 2).

2. The Harmonion Cost ($\mathcal{C}_{\text{harm}}$): The theoretical global minimum cost, obtained by evaluating $\mathcal{C}_p$ on the diagonal form of the Hamiltonian. This corresponds to the cost if the system could be perfectly unscrambled into non-interacting normal modes. Note that $\mathcal{C}_{\text{harm}}$ is a property of the spectrum, while $\mathcal{C}_{\text{spatial}}$ is a property of the interaction graph.

3 Results: The Accessibility Phase Transition

To investigate the emergence of structure, we subjected the unitary orbits of Heisenberg spin systems to the double bracket flow defined in Eq. (9). In all experiments, the system was initialized in a highly non-local "scrambled" state, generated by applying a random unitary circuit of depth $D = N$ to a local seed Hamiltonian. We then tracked the descent of the locality cost $\mathcal{C}_p(H(t))$ until convergence.

3.1 The Accessibility Collapse

We first examine the dependence of structural recovery on system size N, fixing the locality penalty strength at $p = 4$. The results reveal a sharp transition in the accessibility of the global minimum (Table 1).

For micro-scale systems ($N \leq 3$), the optimization landscape permits full recovery of the global minimum. The flow consistently identifies the eigenbasis (Harmonion basis), achieving a cost $\mathcal{C} \approx 1.0$. This indicates that for small Hilbert spaces, the distinction between "spatial" and "spectral" structure is dynamically irrelevant; the system simply slides to the simplest possible description.

In contrast, for $N \geq 5$, we observe an *accessibility collapse*. Despite the mathematical existence of the Harmonion basis (with theoretical cost $\mathcal{C}_{\text{harm}} \approx 4.2$), the optimization flow never reaches it. Instead, the system is dynamically arrested in a local basin of attraction corresponding to the 1D spatial lattice ($\mathcal{C}_{\text{spatial}} = 16.0$).

The divergence between the *accessible* cost and the *optimal* cost grows with system size. This suggests that the "Harmonion" vacuum—a universe of pure, non-interacting modes—is a Platonic ideal that becomes kinetically forbidden in the thermodynamic limit.

Table 1: **The Accessibility Collapse ($p = 4$).** Comparison of the theoretical minimum cost (Eigenbasis) vs. the dynamically recovered cost. At $N = 5$, the system fails to access the global minimum and is trapped by the spatial attractor.

System Size (N)	Spatial Target	Eigenbasis (Ideal)	Recovered Cost	Regime
3	16.0	1.00	**1.01**	Fluid (Global Min)
4	16.0	~ 5.6	**7.4 - 9.1**	Transitional
5	16.0	~ 4.2	**17.1 - 19.8**	Trapped (Spatial)

3.2 Universality of the Spatial Attractor

To verify that this trapping is not an artifact of the 1D ring topology, we repeated the experiments on interaction graphs with varying connectivity, including 2D grids and random regular graphs ($N = 6$).

We find that the accessibility collapse is robust to topology. For $N = 6$, regardless of whether the underlying graph is a ring, a grid, or a random network, the optimization flow fails to find the eigenbasis (Cost $\sim 4 - 6$). In all cases, the recovered cost saturates near the locality cost of the interaction graph itself ($\mathcal{C} \sim 16 - 20$). This universality implies that the emergence of locality is a generic feature of large-N unitary orbits under geometric constraints, rather than a peculiarity of 1D chains.

3.3 The Phase Diagram of Geometry

Finally, we map the phase diagram of accessibility by varying the penalty exponent p for a fixed system size of $N = 5$ (Figure 1). We identify three distinct dynamical phases:

1. **The Quantum Fluid Phase ($p \leq 2$):** In the weak-penalty regime, the cost function lacks sufficient curvature to enforce locality. The flow tunnels through spatial barriers and consistently accesses the delocalized Harmonion basis ($\mathcal{C} \approx 1.2$). In this regime, space cannot exist.

2. **The Emergent Geometry Phase ($p \approx 3 - 4$):** This is the "window of reality." The penalty is strong enough to create high barriers around the non-local eigenbasis, rendering it inaccessible. The system is trapped in the basin of the spatial interaction graph. Here, geometry emerges as the simplest *accessible* structure.

3. **The Glassy Phase ($p \geq 5$):** When the penalty is too harsh, the landscape becomes rugged (glassy). The optimizer gets trapped in high-cost metastable states ($\mathcal{C} \gg 16$), unable to find even the spatial basin. This regime corresponds to structural disorder.

Table 2: **Phase Transition Data ($N = 5$).** The system transitions from a delocalized fluid to a trapped geometric phase as the locality penalty increases.

Penalty (p)	Spatial Target	Recovered Cost	Phase
1.0	2.0	**1.2**	Quantum Fluid
2.0	4.0	**2.1**	Quantum Fluid
3.0	8.0	**6.6**	Geometry (Trapped)
4.0	16.0	**19.7**	Geometry (Trapped)
5.0	32.0	**62.0**	Glassy Disorder

4 The Phase Diagram of Geometry

4.1 Metric Pressure as a Control Parameter

We vary the penalty exponent p to probe the roughness of the optimization landscape.

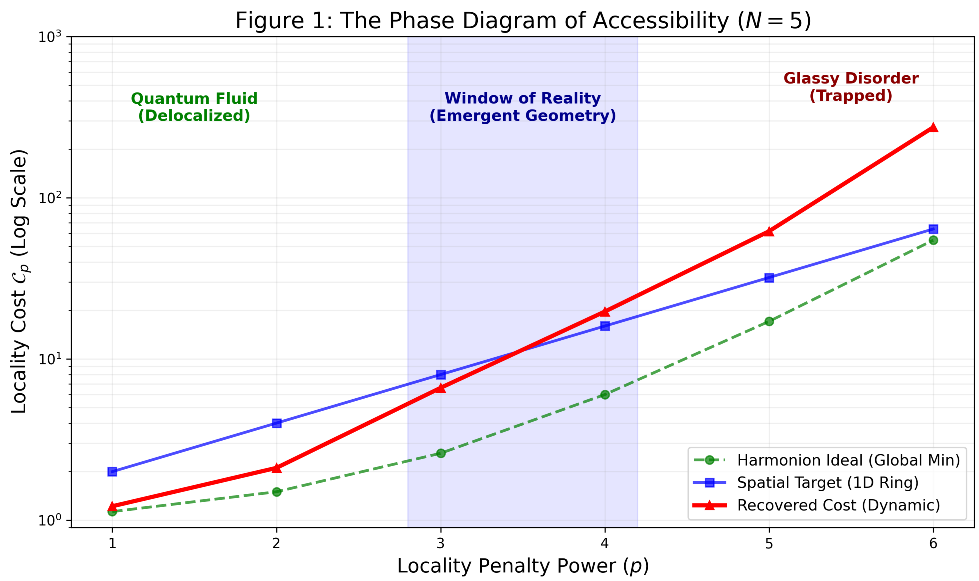

Figure 1: **The Phase Diagram of Accessibility.** The relationship between penalty strength p and structural recovery. The "Window of Reality" ($p \approx 3 - 4$) represents the regime where spatial geometry is the stable attractor. Below this window, the universe dissolves into fluid; above it, it freezes into disorder.

4.2 Three Regimes of Reality

Based on numerical sweeps at $N = 5$, we identify three phases:

1. **Quantum Fluid ($p \leq 2$):** Landscape is smooth; delocalization prevails.
2. **Emergent Geometry ($p \approx 3 - 4$):** Landscape allows trapping in the Spatial basin.
3. **Glassy Disorder ($p \geq 5$):** Landscape is too rugged; system freezes in random configurations.

Penalty Power (p)	Spatial Cost	Recovered Cost	Phase
1.0	2.0	~ 1.2	Fluid (Escaped)
3.0	8.0	~ 6.6	Geometry (Trapped)
6.0	64.0	~ 273	Glass (Lost)

Table 3: Phase transition data for $N = 5$.

5 Discussion

5.1 The Survival of the Robust

The central finding of this work is the distinction between the *existence* of a simple description and its *accessibility*.

Standard reductionism assumes that the universe should relax to its simplest possible configuration—the energy eigenbasis. In our terminology, this is the "Harmonion" vacuum, where all degrees of freedom are independent and non-interacting. Our data for $N = 3$ confirms that this is indeed the ground state of the locality cost landscape.

However, our results for $N \geq 5$ demonstrate that in larger systems, this global minimum becomes kinetically irrelevant. We interpret this through the lens of landscape theory [3]. The Harmonion minimum is deep but extremely narrow—a "golf hole" in the high-dimensional unitary manifold. Finding it requires precise, fine-tuned unscrambling of all global entanglement.

In contrast, the spatial basin is shallow but wide. It represents a class of states that are locally entangled but globally separable. As the system size increases, the entropic probability of finding the narrow channel to the eigenbasis vanishes, and the optimization flow is statistically forced into the broad basin of spatial geometry. Within this model, spatial locality emerges not because it is the simplest structure, but because it is the most robust structure accessible from generic initial conditions under locality-biased flows.

5.2 Connection to Information Propagation

This work complements our previous findings on Heterogeneity in Information Propagation (HIP) [1]. In Paper I, we established that spatial structure is characterized by heterogeneous delays in information transport (light cones).

Here, we provide the dynamical mechanism that selects for that heterogeneity. A universe in the Harmonion basis would have zero HIP; information would not propagate, it would simply phase-rotate in place. By getting trapped in the spatial basin, the system retains a non-zero interaction complexity. This complexity is precisely what allows for the emergence of causal structure, time-delays, and ultimately, local observers.

5.3 Limitations and Future Directions

Our numerical experiments were limited to small system sizes ($N \leq 6$) due to the exponential scaling of the unitary group dimension (4^N). While the phase transition at $N = 5$ is sharp and robust across topologies, verifying this scaling behavior for thermodynamic limits ($N \to \infty$) remains a challenge for future work, potentially utilizing tensor network methods to approximate the manifold geometry.

Additionally, our current model assumes a fixed penalty exponent p. An interesting extension would be to promote p to a dynamical variable, allowing the "metric pressure" to evolve with the system, potentially modeling the cooling of the early universe from a quantum fluid phase into a geometric phase.

6 Conclusion

We have identified an *accessibility phase transition* in the structure of unitary orbits. We demonstrated that for sufficiently large systems ($N \geq 5$) under moderate locality constraints ($p \approx 4$), the theoretically optimal description of the universe (the eigenbasis) becomes dynamically inaccessible.

Consequently, the system spontaneously breaks symmetry into a sub-optimal but kinetically robust configuration: spatial geometry. This suggests that the locality of physics is not a fundamental axiom, but an emergent property—a glassy state of the Hilbert substrate that arises because the universe is too complex to find its own simplest basis.

References

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